

Knowing the Data Before the Model

DEX 2.0 spending and utilisation, 2010–2019

Emily Johnson

31 August 2026 · No models are fit anywhere in this deck

What is in the file

Shape, denominators, and one consequential choice

One file, 29,040 rows, and no gaps

Dimension	Values
Location	51 states + DC, plus a United States row
Year	2010–2019, all ten equally represented
Payer	Medicaid, Medicare, private, out-of-pocket
Type of care	AM, DV, ED, HH, IP, NF, RX
Age basis	<i>All ages</i> and <i>Age/sex-standardized</i>
Cause	Single row, “All causes”

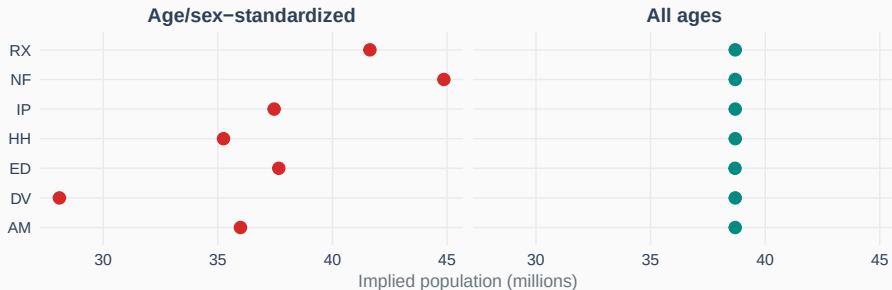
Every state–year–payer has exactly 7 types of care: the panel is perfectly balanced, 2,040 state–year–payer cells. Spending columns have zero missing values.

Five outcome columns, three distinct denominators

Column	Denominator	Reads as
spend_per_capita	state residents	how much a state spends
spend_per_bene	payer enrollees	how much an enrollee costs
vol_per_capita	state residents	care per resident
vol_per_bene	payer enrollees	care per enrollee
spend_per_vol	units of care	the price of a unit

vol_per_bene is per 1,000 enrollees. Out-of-pocket has no enrollee denominator: all 3,640 of its per-beneficiary values are missing by construction, which is every out-of-pocket row in the file.

The age basis decides whether the denominators mean anything



California Medicaid, 2015. Under *All ages* the implied population is 38.69M for every type of care — the true state population. Under *Age/sex-standardized* it ranges from 28M to 45M, because each type of care is reweighted by its own age profile. Standardised rates are comparable across states; they are not a population.

Both denominators are recoverable from the file

$$\text{enrollees} = \frac{\text{spend_mean}}{\text{spend_per_bene}}$$

$$\text{population} = \frac{\text{spend_mean}}{\text{spend_per_capita}}$$

Checked against known values: the recovered 2015 populations are 321.8M for the United States, 38.69M for California, 27.55M for Texas. Recovered enrollee counts agree across all seven types of care within a state-year-payer, confirming a single shared denominator.

The identity

What per-capita spending is made of

Per-capita spending is three things multiplied together

$$\underbrace{\frac{\text{spend}}{\text{pop}}}_{\text{per capita}} = \underbrace{\frac{\text{spend}}{\text{vol}}}_{\text{price}} \times \underbrace{\frac{\text{vol}}{\text{enrollees}}}_{\text{use per enrollee}} \times \underbrace{\frac{\text{enrollees}}{\text{pop}}}_{\text{coverage}}$$

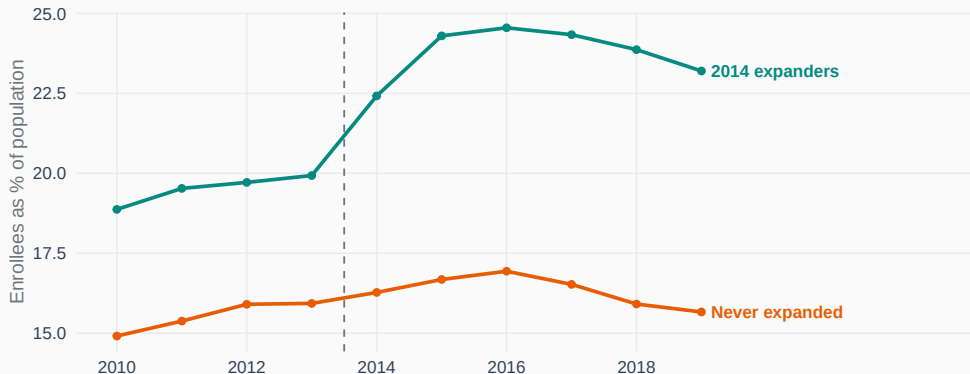
and the middle two collapse to spend per enrollee

An accounting identity, not an assumption. It reconciles exactly under *All ages*: for 2014 expanders in 2016, $\$6,213.5 \times 24.5\% = \$1,525.4$, the reported per-capita figure to the dollar. It does **not** reconcile in the standardised data.

Stepping through

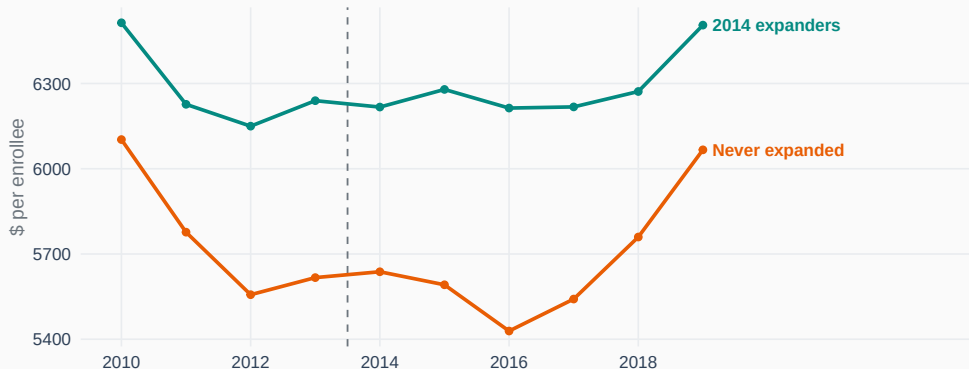
2014 expanders vs. states that never expanded

Coverage: expanders gain 4.5 points, non-expanders gain 0.5



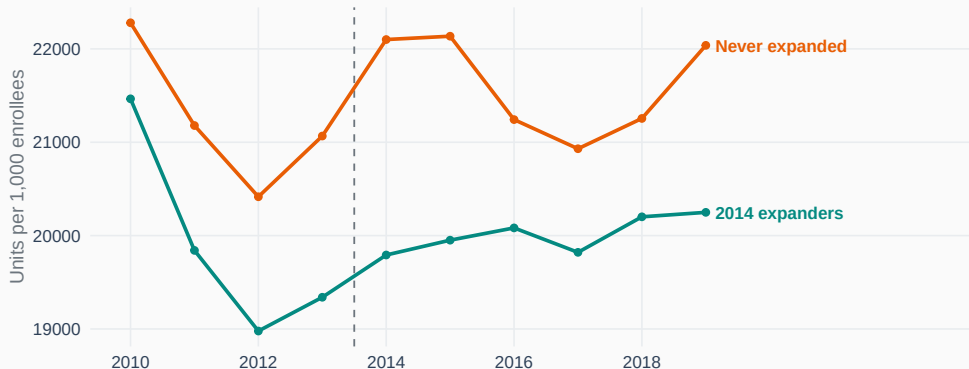
Medicaid enrollees as a share of state population, population-weighted. Dashed line is the last pre-expansion year. 25 expanders vs 19 never-expanders; the 7 later cohorts are held out throughout.

Spend per enrollee: flat in both groups



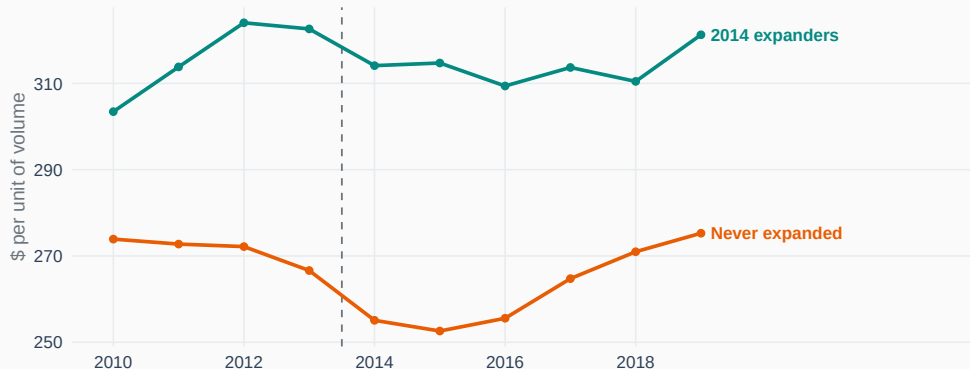
Medicaid spending per enrollee. From 2013 to 2016 this falls 0.4% among expanders and 3.3% among non-expanders. The level gap between the groups is present throughout the pre-period.

Volume per enrollee: a small rise among expanders



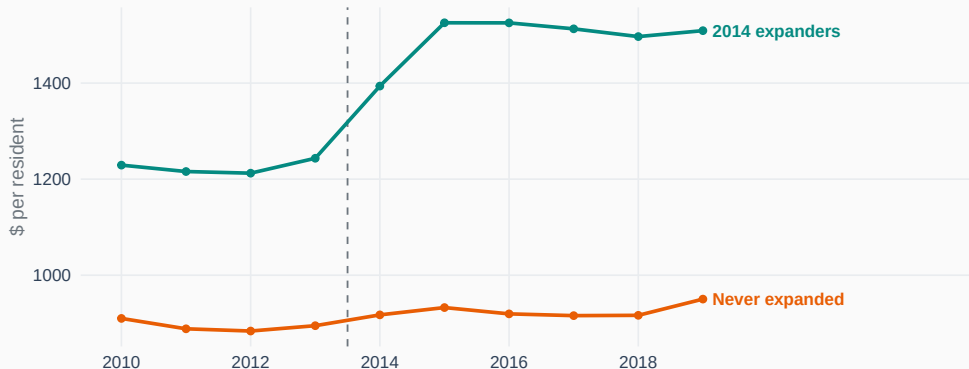
Units of care per 1,000 Medicaid enrollees. From 2013 to 2016, +3.8% among expanders and +0.8% among non-expanders. Note that expanders sit *below* non-expanders in levels for the whole period.

Price per unit: falls 4.1% in both groups



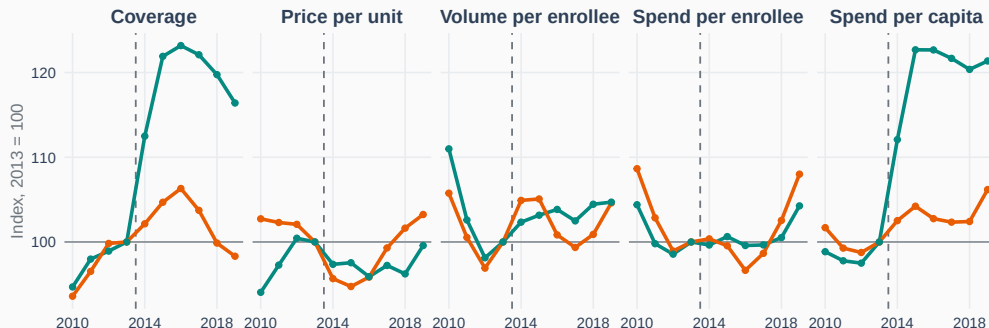
Medicaid spending per unit of volume. The 2013–2016 change is -4.1% in both groups, to one decimal place.

Spend per capita: the two groups separate after 2013



Medicaid spending per state resident. From 2013 to 2016, +22.7% among expanders and +2.8% among non-expanders.

All five components on one scale



Each series indexed to its own 2013 value: **2014 expanders**, **never expanded**. Coverage and spend per capita move together; price, use per enrollee, and spend per enrollee do not separate.

The same numbers, 2013 to 2016

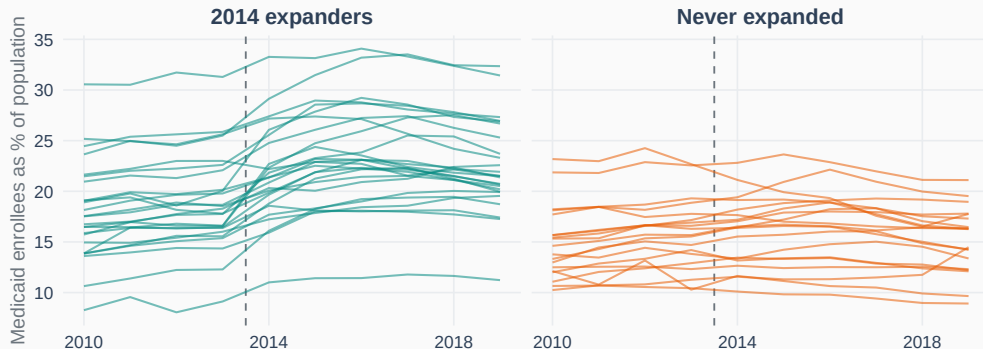
Component	Expanders	Never expanded
Coverage	+23.2%	+6.3%
Price per unit	−4.1%	−4.1%
Volume per enrollee	+3.8%	+0.8%
Spend per enrollee	−0.4%	−3.3%
Spend per capita	+22.7%	+2.8%

Medicaid, all types of care, population-weighted. The bottom row is the product of the coverage row and the spend-per-enrollee row.

Underneath the averages

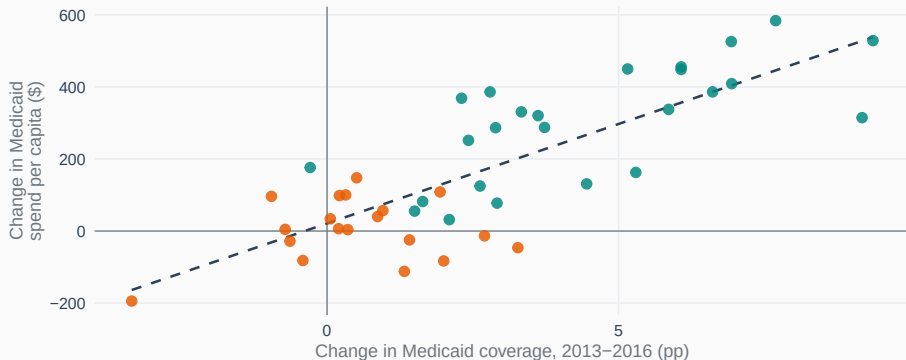
Where the group means come from

Every state's coverage path, not just the group mean



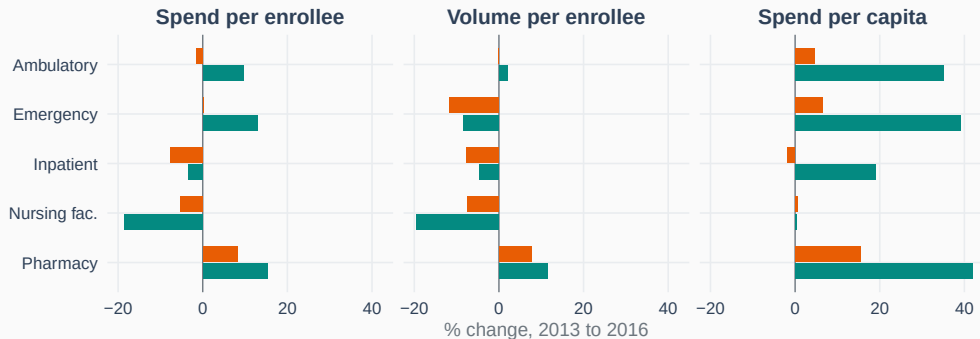
One line per state. The **expander** panel shows a common break at 2014 with varying step size; the **never-expander** panel shows level differences but no common break.

States that gained more coverage gained more spending per capita



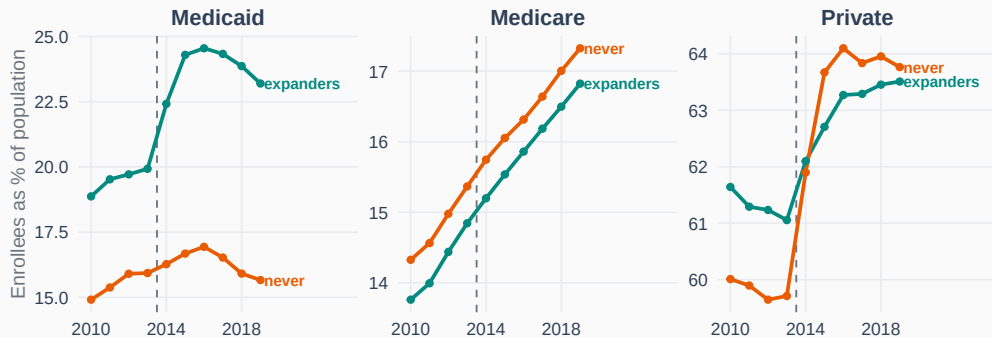
Each point is one state, 2013 to 2016: **2014 expanders**, **never expanded**. $r = 0.80$; the fitted slope is \$55 per resident for each additional coverage point. Expanders average +4.45 points and +\$301; never-expanders average +0.53 points and +\$6.

The pattern by type of care



Percentage change 2013 to 2016, Medicaid; **2014 expanders** vs **never expanded**. Per-enrollee measures are the two left panels; per-capita is on the right. Nursing facility and pharmacy behave differently from ambulatory and inpatient.

Medicare and private coverage move in parallel across both groups



Enrollees as a share of population, by payer. Private coverage rises in both groups after 2013; Medicare rises steadily throughout with no break. Note the free vertical scale on each panel.

Before modelling

Four features of the data to carry forward

Four things the data forces on any model

OPEN **Out-of-pocket has no enrollee denominator.** Per-enrollee outcomes do not exist for it, so it compares only on per-capita and price terms.

OPEN **The age basis is a real choice.** The decomposition holds under *All ages* and fails under *Age/sex-standardized*, which is what the models currently use.

OPEN **Dental and home health exist in the file** but are excluded from the current five types of care, so any “total” is a partial total.

OPEN **Coverage peaks in 2016 and drifts down after.** The post-period is not a plateau, which matters for reading event-study leads.

Every per-capita number in this project is

price × use per enrollee × coverage

Only the third term separates the two groups

Figures: `slides/make_eda_figures.R` · Source: DEX 2.0, *All ages*, 2010–2019